

Worksheet 1 – Solutions

Problem 1(a)[Bernoulli's inequality]

Using the principle of Mathematical Induction, we shall prove

$$(1 + a)^n \geq 1 + na \quad \forall n \geq 2.$$

Solution. Base Case. For $n = 2$, we have

$$(1 + a)^2 = 1 + 2a + a^2 \geq 1 + 2a.$$

Induction hypothesis: Suppose for $k \in \mathbb{N}$, we have

$$(1 + a)^k \geq 1 + ka.$$

Now for $n = k + 1$,

$$\begin{aligned} (1 + a)^{k+1} &= (1 + a)^k(1 + a) \\ &\geq (1 + ka)(1 + a) \quad (\text{Why?}) \\ &= 1 + a + ka + ka^2 \\ &= 1 + (k + 1)a + ka^2 \\ &\geq 1 + (k + 1)a. \end{aligned}$$

This concludes the proof.

Problem 1(b)

$$\lim_{n \rightarrow \infty} c^{1/n} = 1 \quad (c > 0).$$

Case 1: $c = 1$. Trivial.

Case 2: $c > 1$. Then $c^{1/n} = 1 + d_n$ for some $d_n > 0$. Hence, by Bernoulli's inequality,

$$c = (1 + d_n)^n \geq 1 + nd_n.$$

Therefore $c - 1 \geq nd_n$, so that $d_n \leq (c - 1)/n$. Finally,

$$|c^{1/n} - 1| = d_n \leq \frac{c - 1}{n} \quad \text{for } n \in \mathbb{N},$$

which shows that $\lim(c^{1/n}) = 1$ (using the below stated result).

Result: Let (x_n) be a sequence and $x \in \mathbb{R}$. If (a_n) is a sequence of positive real numbers with $\lim(a_n) = 0$ and if there exist $C > 0$ and $n_0 \in \mathbb{N}$ such that

$$|x_n - x| \leq Ca_n \quad \forall n \geq n_0,$$

then $(x_n) \rightarrow x$.

Case 3: $0 < c < 1$. Then $c^{-1/n} = \frac{1}{c^{1/n}} = 1 + h_n$ for $h_n > 0$. Hence, by Bernoulli's inequality,

$$\frac{1}{c} = (1 + h_n)^n \geq 1 + nh_n > nh_n,$$

which shows that $0 < h_n < \frac{1}{nc}$ for $n \in \mathbb{N}$. Therefore

$$0 < 1 - c^{1/n} = \frac{h_n}{1 + h_n} < h_n < \frac{1}{nc}.$$

So that

$$|c^{1/n} - 1| < \left(\frac{1}{c}\right) \frac{1}{n} \quad \text{for } n \in \mathbb{N}.$$

Again using the above stated result in Case 2, we have

$$\lim(c^{1/n}) = 1.$$

Problem 2

Choose the sequence (x_n) where $x_n = (-1)^n$. Then $|x_n| = 1 \rightarrow 1$, but x_n is not convergent.

Problem 3

(i)

$$x_n = \frac{n}{n+1} = \frac{1}{1 + \frac{1}{n}}.$$

Since $(\frac{1}{n}) \rightarrow 0$, we have $(1 + \frac{1}{n}) \rightarrow 1$ and hence

$$(x_n) = \left(\frac{1}{1 + \frac{1}{n}}\right) \rightarrow 1.$$

(ii) $(x_n) = \left((-1)^n \frac{n}{n+1}\right)$ is not convergent. Suppose (x_n) is convergent. Then

$$\left(\frac{x_n(n+1)}{n}\right)$$

is convergent, and hence $((-1)^n)$ is convergent.

Can you identify which result we are using?

(iii) $(x_n) = (\sin n)$ is not convergent. (We need subsequences; I request student to think more about this problem.)

Problem 4

(i) Let $p < 1$. Then for any $n \in \mathbb{N}$ we have $n \geq n^p$ and hence

$$\frac{1}{n} \leq \frac{1}{n^p}.$$

(Now use comparison theorem.)

(ii) Let $p \geq 2$. Then for any $n \in \mathbb{N}$ we have $n^p \geq n^2$ and hence

$$\frac{1}{n^p} \leq \frac{1}{n^2}.$$

Again use comparison theorem.

(iii)

$$\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n^2}\right) = 1 \neq 0 \quad \left(\sum a_n \text{ converges} \Rightarrow (a_n) \rightarrow 0\right).$$

(iv) Apply ratio test. In fact, $\lim_{n \rightarrow \infty} \frac{|(n+1)e^{-(n+1)})|}{|ne^{-n}|} = \lim_{n \rightarrow \infty} \frac{1}{e} \left|1 + \frac{1}{n}\right| = \frac{1}{e} < 1$

Problem 5

Since $(x_n) \rightarrow c$ (say), for $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that for all $n \geq N$ we have

$$|x_n - c| < \frac{\varepsilon}{2}.$$

Now for $n, m \geq N$ we have

$$\begin{aligned} |x_n - x_m| &= |x_n - c + c - x_m| \\ &\leq |x_n - c| + |c - x_m| \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon. \end{aligned}$$

Problem 6

Recall that $\|x\| = \sqrt{\langle x, x \rangle}$. Now for $x, y \in \mathbb{R}^n$,

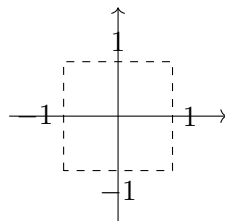
$$\begin{aligned} \|x + y\|^2 + \|x - y\|^2 &= \langle x + y, x + y \rangle + \langle x - y, x - y \rangle \\ &= \langle x, x + y \rangle + \langle y, x + y \rangle + \langle x, x - y \rangle - \langle y, x - y \rangle \\ &= \langle x, x \rangle + \langle x, y \rangle + \langle y, x \rangle + \langle y, y \rangle \\ &\quad + \langle x, x \rangle - \langle x, y \rangle - \langle y, x \rangle + \langle y, y \rangle. \end{aligned}$$

By cancelling the terms $\langle x, y \rangle$ and $\langle y, x \rangle$ we get

$$\|x + y\|^2 + \|x - y\|^2 = 2(\|x\|^2 + \|y\|^2).$$

Problem 7

(i) $|x| < 1$ and $|y| < 1$.



(ii) Now for

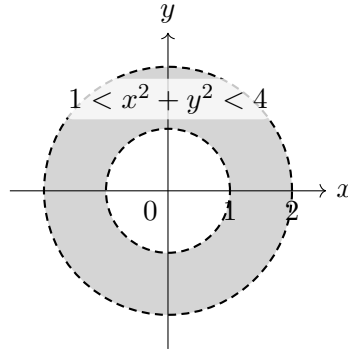
$$(x^2 + y^2 - 1)(x^2 + y^2 - 4) < 0$$

there are two cases:

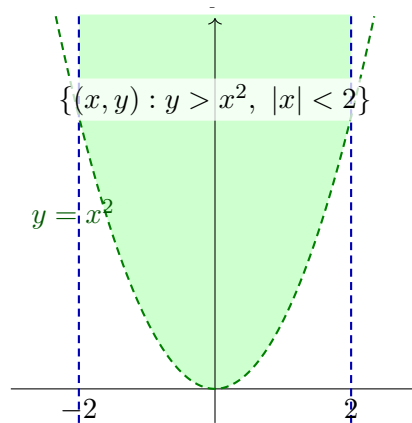
$$\begin{aligned} \text{Case 1: } & x^2 + y^2 - 1 < 0 \quad \text{and} \quad x^2 + y^2 - 4 > 0, \\ \text{or Case 2: } & x^2 + y^2 - 1 > 0 \quad \text{and} \quad x^2 + y^2 - 4 < 0. \end{aligned}$$

Draw these region in the plane? Equivalently, the non-empty common region is

$$1 < x^2 + y^2 < 4.$$



iii $y > x^2$ and $|x| < 2$.



Problem 8

Observe that $\forall n \geq k$,

$$\frac{1}{2n} \leq \frac{1}{n+k} \leq \frac{1}{n}.$$

Thus,

$$\underbrace{\frac{1}{(2n)^2} + \cdots + \frac{1}{(2n)^2}}_{n\text{-times}} \leq \frac{1}{(n+1)^2} + \frac{1}{(n+2)^2} + \cdots + \frac{1}{(2n)^2} \leq \underbrace{\left(\frac{1}{n^2} + \frac{1}{n^2} + \cdots + \frac{1}{n^2} \right)}_{n\text{-times}},$$

which in turn shows that

$$\frac{1}{4n} = \frac{n}{4n^2} \leq \frac{1}{(n+1)^2} + \frac{1}{(n+2)^2} + \cdots + \frac{1}{(2n)^2} \leq \frac{n}{n^2} = \frac{1}{n}.$$

Apply squeeze theorem for sequence.

Problem 9

(a) Observe that $\ln(n+1) > 1$ for sufficiently large n . Now apply the comparison theorem to conclude,

$$\sum \frac{\ln(n+1)}{n+1} \text{ diverges.}$$

(b) Recall that range of $\sec^{-1}(x)$ is $[0, \pi] - \frac{\pi}{2}$ and hence $\sec^{-1} n < \pi$,

$$\frac{\sec^{-1} n}{n^{1.3}} < \pi \frac{1}{n^{1.3}}.$$

Now apply the comparison test to $\sum \frac{1}{n^{1.3}}$ (p-series).

(c)

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{3^{n-1} + 1}{3^n} &= \lim_{n \rightarrow \infty} \left(\frac{3^{n-1}}{3^n} + \frac{1}{3^n} \right) \\ &= \lim_{n \rightarrow \infty} \left(\frac{1}{3} + \frac{1}{3^n} \right) = \frac{1}{3} \neq 0. \end{aligned}$$

Problem 10

(a) Prove that

$$\sum \frac{(-1)^n}{1 + \sqrt{n}}$$

converges conditionally (using alternating series test) but $\sum \frac{1}{1+\sqrt{n}}$ diverges. Since $1 + \sqrt{n} \leq \sqrt{n} + \sqrt{n}$ and hence

$$\frac{1}{1 + \sqrt{n}} \geq \frac{1}{2\sqrt{n}}$$

(apply comparison test).

(b) Converges conditionally (using alternating series test) but $\sum (\sqrt{n+1} - \sqrt{n})$ diverges.

(c) Apply root test to show that

$$\sum \left| \left(\frac{n+1}{2n} \right)^n \right|$$

converges and hence $\sum (-1)^n \left(\frac{n+1}{2n} \right)^n$ is absolutely convergent.

(d) Consider,

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{(2n)!}{2^n n! n} &= \lim_{n \rightarrow \infty} \frac{(n!)(n+1)(n+2) \cdots (2n-1)(2n)}{2^n n! n} \\ &= \lim_{n \rightarrow \infty} \frac{(n+1)(n+2) \cdots (2n-1)}{2^{n-1}} \\ &= \lim_{n \rightarrow \infty} \left(\frac{n+1}{2} \right) \left(\frac{n+2}{2} \right) \cdots \left(\frac{n+(n-1)}{2} \right) \\ &= \lim_{n \rightarrow \infty} \left(\frac{n+1}{2} \right)^{n-1} \\ &= \infty \end{aligned}$$

Now use the n -th term test to conclude that the series $\sum \frac{(2n)!}{2^n n! n}$ is not convergent.